

DATA SCIENCE WITH **AI**

Advanced Data Science
Program with AI

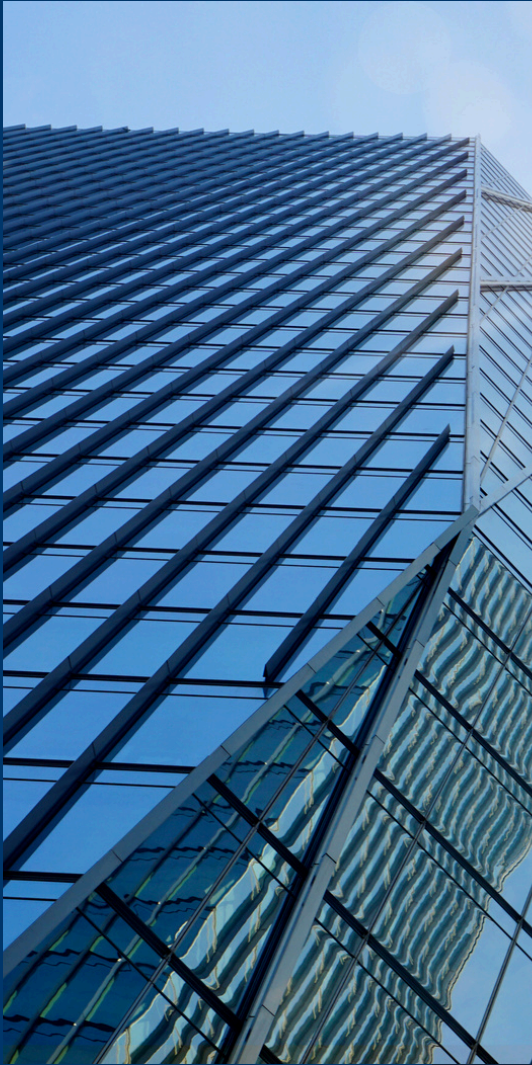


ABOUT DIGITAL GHURU

Digital Ghuru is a forward-thinking digital training and solutions platform dedicated to empowering individuals and businesses with the skills required to succeed in today's digital economy.

With a strong focus on practical learning and real-world application, Digital Ghuru bridges the gap between theoretical knowledge and industry demands. Our programs are designed to help learners build job-ready skills, create sustainable income opportunities, and enable businesses to scale through digital transformation.

We believe that digital education should be accessible, simplified, and outcome driven, enabling every learner to confidently step into the digital world.



ABOUT THE PROGRAM



- Data Scientist
- Data Analyst
- Machine Learning Engineer
- Business Intelligence Analyst

The Advanced Data Science Program with Python is designed to help learners master data science from basics to advanced level.

This program equips you with the ability to analyze data, build machine learning models, and generate business insights using real-world datasets.

After completing the course, you can confidently apply for roles such as:



Why Choose Digital Ghuru ?

Empower your future with practical skills, expert mentorship, real-world experience, and career-focused training designed to help you succeed confidently.



Live Interactive Classes



Industry Expert Trainers



Hands-on Real-Time Projects



AI Tools & Industry Exposure



Placement Guidance & Career Support



Beginner-Friendly Curriculum



Flexible Learning Options



PROGRAM HIGHLIGHTS

- Step-by-step structured learning
- Practical assignments for every module
- Real-world case studies
- Capstone projects for portfolio
- Continuous mentorship support
- Career guidance sessions

TOP SKILLS YOU WILL LEARN

- Python Programming
- Data Analysis & Data Handling
- Data Visualization & Storytelling
- SQL & Database Management
- Machine Learning Models
- Deep Learning Fundamentals
- Cloud & Model Deployment
- Business Problem Solving



TOOLS AND TECHNOLOGIES COVERED

PYTHON AND AWS



PANDAS & NUMPY



MATPLOTLIB & SEABORN



SQL AND SCIKIT-LEARN



TENSORFLOW / PYTORCH



TABLEAU / POWER BI



COURSE CURRICULUM

Structured learning path covering essential concepts, tools, and real-world applications to help you become a confident and job-ready data professional.



MODULE – 1:

PYTHON FUNDAMENTALS

Build a strong foundation in Python by learning core programming concepts including variables, data types, loops, and functions, enabling you to write efficient code for data science applications.

Python Basics

- 1.Introduction to Python
- 2.Variables & Data Types
- 3.Conditional Statements
- 4.Loops (For & While)
- 5.Functions

MODULE – 2:

ADVANCED PYTHON

Enhance your Python skills by exploring advanced concepts like object-oriented programming, file handling, debugging, and regular expressions, preparing you to handle real-world data science coding challenges efficiently.

Advanced Concepts

- 1.Object-Oriented Programming
- 2.File Handling
- 3.Regular Expressions
- 4.Debugging & Exception Handling
- 5.Libraries & Packages



DATA SCIENCE MODULE



MODULE – 3: ALGORITHMIC THINKING

Develop problem-solving skills by understanding algorithms, time complexity, and data structures, enabling you to write optimized, scalable, and efficient solutions for complex computational and data processing problems.

Logic Building

1. Algorithms & Complexity
2. Searching & Sorting
3. Data Structures
4. Best Practices

MODULE – 4: DATA

HANDLING WITH PANDAS

Learn to work with large datasets using Pandas by performing data cleaning, transformation, and manipulation, helping you extract meaningful insights and prepare data for analysis and machine learning tasks.

Data Processing

1. Data Frames & Series
2. Data Cleaning
3. Data Transformation
4. Aggregation & Grouping





MODULE – 5:

SQL FUNDAMENTALS

Understand the basics of databases and SQL by learning how to create, retrieve, update, and manage structured data efficiently, forming a strong base for data analysis and backend operations.

Database Basics

- 1.Tables & Data Types
- 2.CRUD Operations
- 3.SQL Queries
- 4.Constraints

MODULE – 6:

ADVANCED SQL

Master advanced SQL concepts including joins, subqueries, functions, and data integration techniques, enabling you to handle complex queries and efficiently manage large-scale structured datasets in real-world applications.

Advanced Queries

- 1.Joins
- 2.Subqueries
- 3.Functions
- 4.Python Integration



DATA SCIENCE MODULE



MODULE – 7: PROBABILITY & STATISTICS – I

Build a solid statistical foundation by understanding probability concepts, distributions, random variables, and sampling techniques, which are essential for analyzing data and building accurate machine learning models.

Statistics Basics

1. Probability
2. Distributions
3. Expectation & Variance
4. Sampling

MODULE – 8: PROBABILITY & STATISTICS – II

Learn advanced statistical methods including hypothesis testing, confidence intervals, and inferential statistics, helping you draw meaningful conclusions from data and support decision-making with statistical evidence.

Advanced Statistics

1. Hypothesis Testing
2. Central Limit Theorem
3. Confidence Intervals
4. Estimation





MODULE – 9:

DATA VISUALIZATION

Understand how to represent data visually using charts and graphs, enabling you to communicate insights effectively through tools like Matplotlib and Seaborn, improving storytelling and business decision-making capabilities.

Visualization

- 1.Charts & Graphs
2. Matplotlib & Seaborn
3. Data Storytelling

MODULE – 10: DATA

PREPROCESSING

Prepare raw data for analysis by handling missing values, encoding categorical data, detecting outliers, and transforming datasets, ensuring clean, structured, and high-quality input for machine learning models.

Data Cleaning

1. Missing Values
2. Encoding
3. Outliers
4. Feature Preparation



DATA SCIENCE MODULE



MODULE – 11: EXPLORATORY DATA ANALYSIS

Analyze datasets to discover patterns, relationships, and trends using statistical techniques and visualization methods, helping you gain deeper insights and make data-driven decisions effectively in real-world scenarios.

EDA

1. Univariate & Bivariate Analysis
2. Correlation
3. Case Study

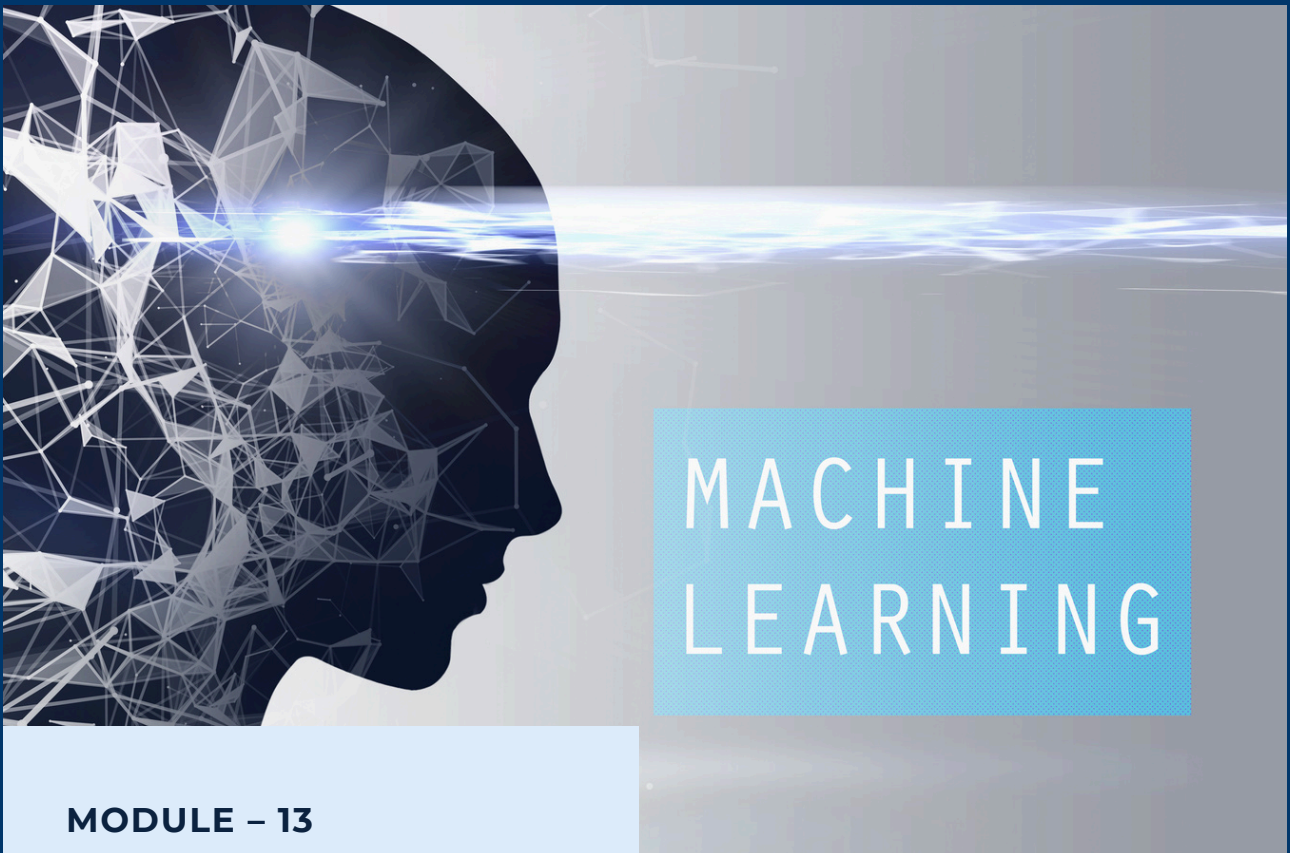
MODULE – 12: MACHINE LEARNING FOUNDATIONS

Understand the fundamentals of machine learning and learn how to build predictive models using regression techniques.

Machine Learning – I

1. Introduction to Machine Learning
2. Supervised Learning Basics
3. Regression Models
4. Model Evaluation Metrics
5. Feature Selection





MODULE – 13

Learn classification techniques used to categorize data and solve real-world prediction problems.

Machine Learning – II

1. Classification Basics
2. Logistic Regression
3. K-Nearest Neighbors (KNN)
4. Support Vector Machine (SVM)
5. Evaluation Metrics

MODULE – 14

Explore tree-based models and ensemble techniques for improving model performance and accuracy.

Machine Learning – III

1. Decision Trees
2. Random Forest
3. Ensemble Methods
4. Boosting Techniques
5. Model Optimization



DATA SCIENCE MODULE



MODULE – 15

Learn advanced techniques to improve model performance and handle real-world data challenges.

Machine Learning – IV

1. Model Comparison
2. Overfitting & Underfitting
3. Imbalanced Data Handling
4. Sampling Techniques
5. Model Improvement

MODULE – 16:

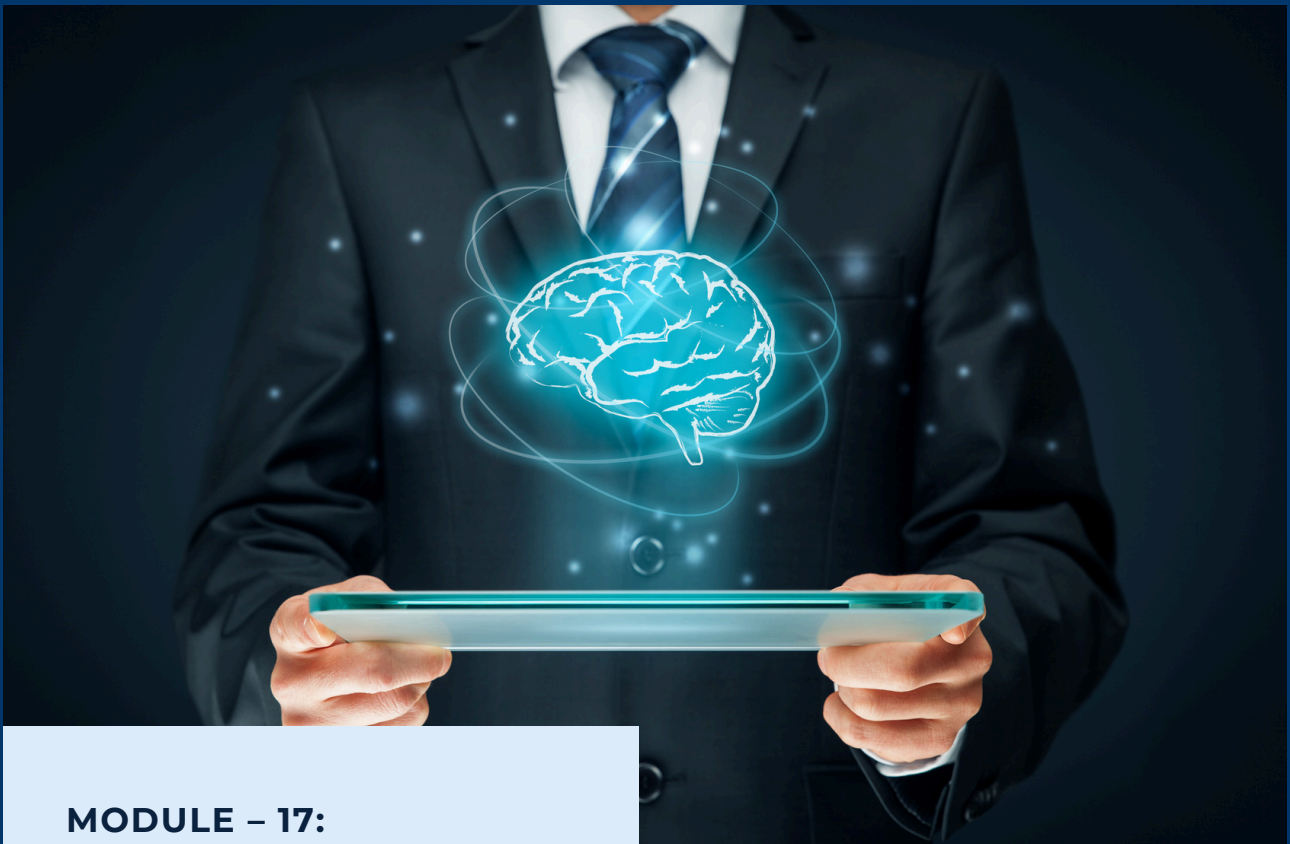
UNSUPERVISED LEARNING

Understand how to identify hidden patterns in data using clustering techniques like K-Means, enabling you to work with unlabeled datasets and solve real-world business problems effectively

Clustering

1. K-Means
2. Use Cases





MODULE – 17:

DEEP LEARNING

Learn advanced concepts of deep learning to handle complex and high-dimensional data using neural networks inspired by the human brain.

Deep Learning

1. Neural Networks Basics
2. Activation Functions (ReLU, Sigmoid, Tanh)
3. Gradient Descent
4. Backpropagation
5. Overfitting & Prevention

MODULE – 18:

DEEP LEARNING

Implement deep learning models using PyTorch and understand the fundamentals of building and training neural networks.

Deep Learning with PyTorch

1. PyTorch Setup & Tensors
2. Autograd & Backpropagation
3. Training Pipeline
4. Regression Models
5. Dataset & Dataloader



DATA SCIENCE MODULE



MODULE – 19

Explore advanced deep learning techniques and architectures used in real-world applications.

Deep Learning with PyTorch – Continued

1. Neural Network Implementation
2. CNN Basics
3. Image Processing
4. Transfer Learning
5. Model Saving & Loading

MODULE – 20: NATURAL LANGUAGE PROCESSING

Learn to process and analyze text data using NLP techniques such as sentiment analysis and text classification, enabling you to build applications like chatbots and language-based AI systems.

NLP

1. Text Processing
2. Sentiment Analysis
3. Chatbots





MODULE – 21:

COMPUTER VISION

Understand how machines interpret visual data by learning image processing, convolutional neural networks, and object detection techniques, enabling development of applications in image recognition and analysis.

Computer Vision

1. Image Processing
2. Object Detection
3. CNN Models

MODULE – 22: MODEL

DEPLOYMENT IN AWS CLOUD

Learn how to deploy machine learning models into production using AWS services, enabling you to build scalable, real-world applications and understand cloud-based infrastructure for data science solutions.

Deep Learning with PyTorch

1. Introduction to AWS
2. Cloud Services (EC2, Lambda, S3, RDS etc)
3. Hands-on in EC2 Instance
4. Hands-on in AWS Database



DATA SCIENCE MODULE



PROJECTS FOR YOUR PORTFOLIO

1. Movie Revenue Prediction
2. Financial Fraud Detection
3. Customer Segmentation
4. NLP-based Chatbot

MODULE – 23:

CAPSTONE PROJECTS

Apply all learned concepts by working on real-world projects, enabling you to build a strong portfolio, gain practical experience, and demonstrate your data science skills to potential employers.

Projects

1. Fraud Detection System
2. Customer Segmentation
3. Recommendation Engine
4. AI Chatbot



PLACEMENT SUPPORT

01

Resume
Building

02

LinkedIn Profile
Optimization

03

Interview
Preparation

04

Career
Guidance

WHO CAN JOIN?



Students &
Freshers



Working
Professionals



Career
Switchers



Entrepreneurs



Digital Ghuru - AI Career Transformation Academy
(A Unit of Jai Ghurudeva Academy Private Limited)

Contact us

-  8825948859
-  info@gmail.com
-  www.digitalghuru.com
-  Anna Nagar | Nungambakkam |
Korattur | Hyderabad

